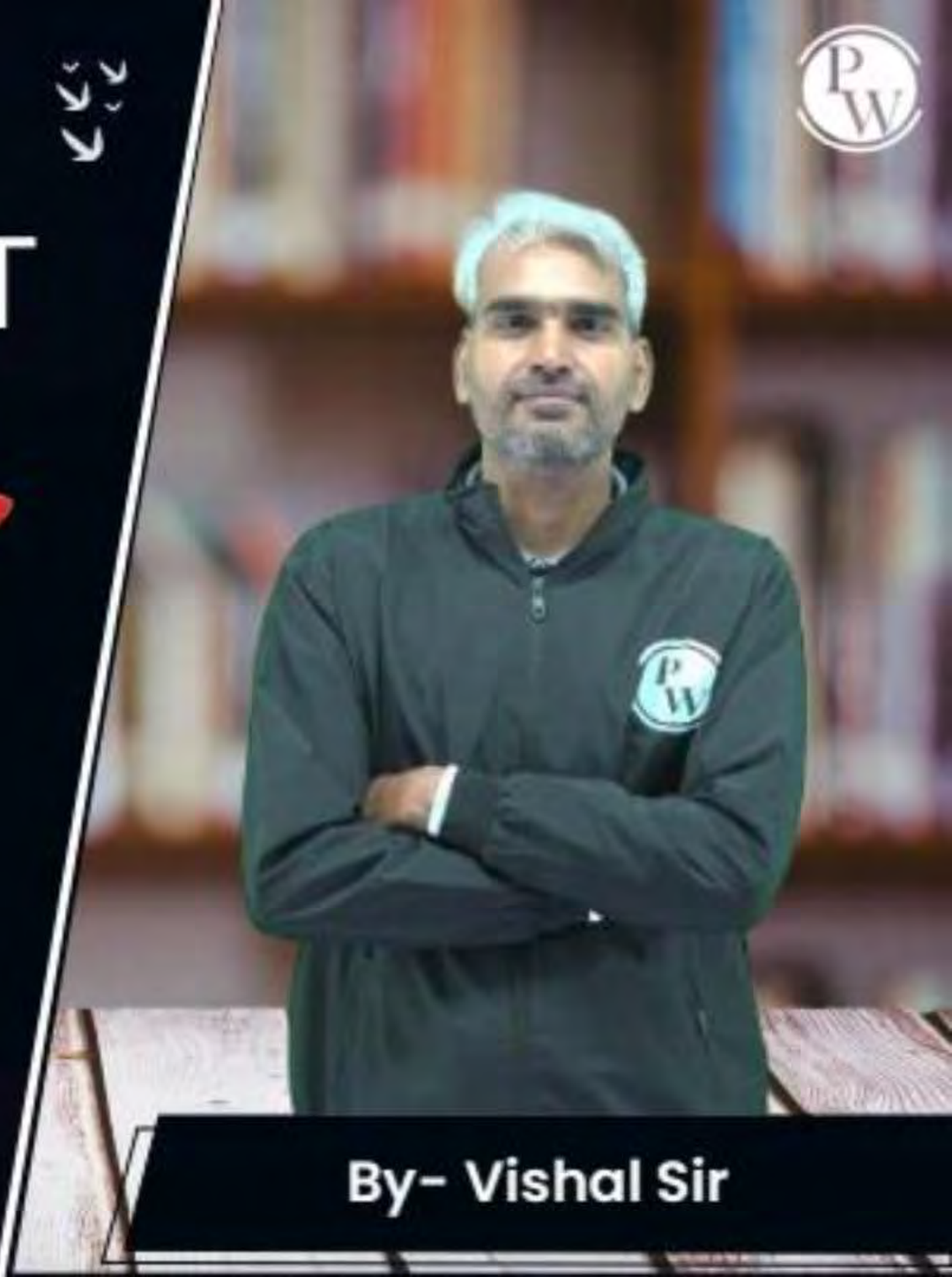
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 22



By- Vishal Sir

Recap of Previous Lecture



Topic Subgroup

Topic Cyclic group

↓

Let $(G,*)$ be a group
if $a \in G$ s.t.
different elements of set G
can be written in the form $(a)^n$ for different positive integers n
then $(G,*)$ is a cyclic group

$(G,*)$ is a group
and $H \subseteq G$, such that
$(H,*)$ is a group
then it is a subgroup of group $(G,*)$

Slide

Topics to be Covered



- **Topic** Cyclic group
- **Topic** Properties w.r.t. cyclic groups
- **Topic** Some important properties

Slide



Topic : Subgroup

* Let $(G, *)$ is a group . a subset H of set G is called a sub-group of group $(G, *)$ if and only if $(H, *)$ is a group.

* Let $(G, *)$ be a group and 'e' is the identity element w.r.t. binary op^n '$*$' then $(G, *)$ and $(\{e\}, *)$ are called trivial sub-group of group $(G, *)$ and any other sub-group of group $(G, *)$ will be called Proper sub group of group $(G, *)$

- Let $(G, *)$ is a finite group of order $= |G|$

then $(G, *)$ is a sub-group of group $(G, *)$ and its order is $|G|$

& $(\{e\}, *)$ is a subgroup of group $(G, *)$ and its order is '1'.

- We know $\{1, -1, i, -i\}$ is a group of order = 4 w.r.t. multiplication

. $\{1\}$ will be a trivial subgroup of order = 1,

$\{1, -1, i, -i\}$ will be a trivial sub-group of order = 4

$\{1, -1\}$ is a proper subgroup of above group. and its order is '2'.

$\{1, i\}$ is a sub-set of set $\{1, -1, i, -i\}$, but it is not a subgroup of given group.

because $\{1, i\}$ is not a group w.r.t. binary op^n multiplication

→ $\{1, 3, 5, 7\}$ is a group of order = 4, w.r.t. $\otimes_8$

- $\{1\}$ w.r.t $\otimes_8$ is a group, } $\{1\}$ & $\{1, 3, 5, 7\}$ are
  $\{1,3,5,7\}$ w.r.t $\otimes_8$ is a group. } trivial sub groups of above group.

→ $\{1,3\}$, $\{1,5\}$ & $\{1,7\}$ are proper sub-groups of above group



Topic : Properties w.r.t. Subgroup

1. Let $(G,*)$ be a group. and H is a non-empty subset of G. then

$(H,*)$ is a sub-group of group $(G,*)$ if and only if:

$$a * b^{-1} \in H, \quad \forall a, b \in H$$

↳ i.e. H is ① Closed
② identity exist
& ③ Inverse exist for every element



Topic : Properties w.r.t. Subgroup

(2) Let $(G,*)$ be a finite group of order $= |G|$, and $(H,*)$ is a sub-group of group $(G,*)$ and order of sub-group $(H,*)$ is $|H|$, then $|H|$ divides $|G|$.

Lagrange's theorem

i.e. order of sub-group divides the order of original group



Topic : Properties w.r.t. Subgroup

③ Let $(G,*)$ be a group, and $(H_1,*)$ & $(H_2,*)$ are any two sub-groups of group $(G,*)$

then $(H_1 \cap H, *)$ is also a sub-group of group $(G,*)$

* Let $a, b \in H_1 \cap H_2$.

$\therefore a, b \in H_1$ and $a, b \in H_2$

we know $(H_1, *)$ is a group

$\therefore a * b^{-1} \in H_1$

we know H_2 is also a group

$\therefore a * b^{-1} \in H_2$

$\therefore a * b^{-1} \in H_1 \cap H_2$

i.e. if $a, b \in H_1 \cap H_2$ then $a * b^{-1} \in H_1 \cap H_2$

$\therefore (H_1 \cap H_2, *)$ is also a group. $\therefore$ it is also a subgroup of $(G, *)$



Topic : Properties w.r.t. Subgroup

④ Let $(G,*)$ be a group and $(H_1,*)$ & $(H_2,*)$ are any two sub-groups of $(G,*)$, then

$(H_1 \cup H_2, *)$ is a subgroup of $(G,*)$ if and only if

① $H_1 \subseteq H_2$ {i.e. $H_1 \cup H_2 = H_2$}

or ② $H_2 \subseteq H_1$ {i.e. $H_1 \cup H_2 = H_1$}

eg We know $\{0, \pm1, \pm2, \pm3, \pm4, \pm5, \ldots\}$ is a group w.r.t. binary opn addition

Let $H_1 = \{0, \pm2, \pm4, \pm6, \pm8, \ldots\}$ is a subset of above set and it is a group w.r.t addition, $\therefore (H_1, +)$ is a subgroup of above group

& $H_2 = \{0, \pm3, \pm6, \pm9, \pm12, \ldots\}$ is also a subset of above set, and it is also a group w.r.t. addition, $\therefore (H_2, +)$ is also a subgroup of above group

$$H_1 \cup H_2 = \{0, \pm2, \pm3, \pm4, \pm6, \pm8, \pm9, \ldots\}$$

$2 + 3 = 5 \notin H_1 \cup H_2$, it is not closed w.r.t. addition

$\therefore H_1 \cup H_2$ is not a group w.r.t. addition

Hence, $(H_1 \cup H_2, *)$ is not a subgroup of given group.

Q:- Let $(G,*)$ be a group of order $= |G|$, if $|G|$ is a prime number, then find the total number of subgroups of group $(G,*)$

- Order of subgroup of group $(G,*)$ must divide order of group i.e. $|G|$

  let $|G| = p$ (Prime no.)

- Because 'p' is a prime number,

  ∴ Order of subgroup can be 1 or p only.

  - 1 → it is w.r.t. trivial sub-group $(\{e\}, *)$
  - p → it is w.r.t. trivial sub-group $(G,*)$

∴ Only two sub-groups are possible, and both of them are trivial

Q: Let $(G, *)$ be a group of order = 8;
then How many sub-groups are possible for group $(G, *)$

↓

Can not be answered,
until we know the elements of the group
and binary opⁿ '$*$'!



Topic : Cyclic group

Let $(G,*)$ be a group, If these exist any element $a \in G$, such that every element of set G can be written in the form $(a)^n$ for some positive integer 'n', then group $(G,*)$ is called a cyclic group, and element 'a' is called generator w.r.t. cyclic group $(G,*)$.

$$(a)^n = \underbrace{a*a*a*\cdots*a}_{n \text{ times } a}$$

A cyclic group may have more than one generator

eg. We know $\{1, -1\}$ is a group of order = 2, w.r.t multiplication

↳ $O(G) = 2$

1 = identity

$(1)^1 = 1$ = identity

$(1)^2 = 1$ = identity

⋮

$(e)^1 = e$

$(e)^2 = e$

⋮

$(e)^n = e$

→ identity element can not generate any other element except itself.

$(-1)^1 = -1$ }

$(-1)^2 = 1 = e$ } ← '-1' can generate all elements of the set for different powers of '-1'

$\therefore O(-1) = 2 = O(G)$

$\Rightarrow$ $\therefore$ $\{1, -1\}$ is a cyclic group w.r.t. multiplication, with '-1' as its generator

eg: We know $\{1, \omega, \omega^2\}$ form a group of order = 3, w.r.t. multiplication

- 1 = identity ∴ '1' can not generate any other element except itself

- $(\omega)^1 = \omega$
  $(\omega)^2 = \omega^2$
  $(\omega)^3 = \omega^3 = 1 = e$

  ← 'ω' can generate all the elements ∴ $\{1, \omega, \omega^2\}$ is a cyclic group & 'ω' is one of its generator

$O(\omega) = 3 = O(G)$

$(\omega^2)^1 = \omega^2$
$(\omega^2)^2 = \omega$
$(\omega^2)^3 = 1 = e$

← ω^2 can also generate all elements.

$O(\omega^2) = 3 = O(G)$

∴ ω^2 is also the generator of Cyclic group $\{1, \omega, \omega^2\}$ w.r.t. multiplication

$\text{inv}(\omega) = \omega^2$

∴ ω^2 will also be a generator

eg: We know $\{1, -1, i, -i\}$ is a group of order = 4 wrt multiplication

$O(G) = 4$

* 1 = identity. ∴ can not generate any other element except itself

* $(-1)^1 = -1$
  $(-1)^2 = 1 = e$
  $(-1)^3 = -1$
  $(-1)^4 = 1 = e$
  $(-1)^5 = -1$
  $(-1)^6 = 1 = e$

elements are repeated

Once we have obtained the identity element, if we increase the power then already generated elements will be repeated

$(i)^1 = i$
$(i)^2 = -1$
$(i)^3 = -i$
$(i)^4 = 1 = e$

$O(i) = 4 = O(G)$

$\therefore$ 'i' is a generator

$\text{inv}(i) = -i$
$\therefore$ '$-i$' will also be a generator

all elements

$(-i)^1 = -i$
$(-i)^2 = -1$
$(-i)^3 = +i$
$(-i)^4 = 1 = e$

$O(-i) = 4 = O(G)$

$\therefore$ '$-i$' is also a generator

all elements $\therefore$ 'i' is a generator

$(1, -1, i, -i)$ is a cyclic group

→ i.e. generator needs to generate all other elements of the set before it generates identity element

Q: $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a group of order $= 4$

$\hookrightarrow O(G) = 4$

* 1 = identity, i.e. '1' can not generate any other element except itself.

* $(3)^1 = 3$

  $(3)^2 = 3 \otimes_8 3 = 1 = e$ ← does not generate all elements of the group.

  $\boxed{O(3) = 2 \neq O(G)}$

* $(5)^1 = 5$

  $(5)^2 = 5 \otimes_8 5 = 1 = e$ ← does not generate all elements of the group.

  $\boxed{O(5) = 2 \neq O(G)}$

$(7)^1 = 7$

$(7)^2 = 7 \otimes_8 7 = 1 = e$ ← does not generate all elements

$\boxed{O(7) = 2 \neq O(G)}$

No element of the group $\{1, 3, 5, 7\}$ w.r.t $\otimes_8$ can generate all elements of the group.

∴ generator of the elements of the group does not exist.

Hence, group is not a cyclic group.



Topic : Cyclic group

① Identity element can not be the generator of a set containing any other element except itself.



Topic : Cyclic group

② Let $(G, *)$ be a finite group of order $= |G|$.

if there exist any element $a \in G$, such that

$O(a) = O(G)$, then $(G, *)$ is a cyclic group

and element 'a' is one of

the generator of cyclic group $(G, *)$



Topic : Cyclic group

③ Let $(G,*)$ be a finite group of order $= |G|$.

if there exist no element $a \in G$, such that

$O(a) = O(G)$, then $(G,*)$ is not a cyclic group.

i.e

$O(a) \neq O(G)$, $\forall a \in G$



Topic : Cyclic group

(1) Let $(G, *)$ be a finite group of order $= |G|$.

if there exist any element $a \in G$, such that

$O(a) = O(G)$, then $(G, *)$ is a cyclic group and element 'a' is one of the generator of cyclic group $(G, *)$

We know $O(a) = O(a^{-1})$

$\therefore$ if $O(a) = O(G)$, then $O(a^{-1}) = O(G)$

Hence if element $a \in G$ is a generator, then a^{-1} is also a generator of that cyclic group.

Q: We know $\{0, 1, 2, 3, 4\}$ is a group of order = 5 (Prime No.) w.r.t. binary opn $\oplus_5$,

find all generators of above group

Soln: Order of group is = 5 = prime no.

∴ order of elements will divide order of group.

i.e., order of elements will be 1 or 5

1 for identity element → it can never be the generator containing any other element except itself.

5 for every other non-identity element

∴ $O(1) = 5 = O(G)$ ∴ generator

$O(2) = 5 = O(G)$ ∴ generator

$O(3) = 5 = O(G)$ ∴ generator

$O(4) = 5 = O(G)$ ∴ generator

Q: Let $\{1, 2, 3, 4, 5, 6\}$ is a group w.r.t. $\otimes_7$.

Check whether group is a cyclic group or not?

If Cyclic then identify all generators of the group.

$O(G) = 6$ = not a prime number

1 = identity,

$\therefore O(1) = 1 \neq O(G)$

$\therefore$ '1' can not be the generator

$(2)^1 = 2$

$(2)^2 = 4$

$(2)^3 = 1$ = identity

$O(2) = 3 \neq O(G)$ $\therefore$ '2' is not a generator

$\text{inv}(2) = 4$ $\therefore$ 4 is not a generator

$(3)^1 = 3$

$(3)^2 = 2$

$(3)^3 = 6$

$(3)^4 = 4$

$(3)^5 = 5$

$(3)^6 = 1$ = identity

$O(3) = 6 = O(G)$

$\therefore$ 3 is a generator

$\text{inv}(3) = 5$

$\therefore$ 5 is also a generator

$(6)^1 = 6$

$(6)^2 = 1$ = identity

$O(6) = 2 \neq O(G)$

$\therefore$ 6 is not a generator

$\text{inv}(6) = 6$



Topic : Properties w.r.t. Cyclic group

Let $(G,*)$ be a Cyclic group with $O(G) = n$, and element $a \in G$ is one of the generators of cyclic group $(G,*)$, then,

$\{ a^m \mid m \in$ Natural no. and $m < n$ and $GCD(m,n) = 1 \}$

is a set of all generators of cyclic group $(G,*)$

$a^m = \underbrace{a * a * \cdots * a}_{m \text{ times } a}$

Q: Let $\{0, 1, 2, 3, 4\}$ is a cyclic group w.r.t. $\oplus_5$, and '1' is one of the generators of the given cyclic group. Find all the generators of the above cyclic group.

Soluⁿ $O(G) = n = 5$, & $\boxed{a = 1}$

Natural numbers that are less than 5 & Co-prime to 5 are

1, 2, 3, 4

∴ The set of all generators of above cyclic group is

$= \{(a)^1, (a)^2, (a)^3, (a)^4\}$

$= \{(1)^1, (1)^2, (1)^3, (1)^4\}$

$= \{1, 2, 3, 4\}$



Topic : Properties w.r.t. cyclic group

- Let $(G,*)$ be a cyclic group of order "$O(G)=n$". then number of generators of cyclic group $(G,*)$ will be equal to the number of positive integers less than n and Co-prime to 'n'

Q: Let $\{0, 1, 2, 3, 4\}$ is a cyclic group w.r.t. $\oplus_5$, then find the number of generators of the above cyclic group.

Soln: for the given cyclic group $O(G) = 5$

∴ No. of generators = No. of +ve integers less than 5 and Co-prime to '5'

1, 2, 3 & 4 = 4

One such numbers

Q. Let $(G, *)$ be a cyclic group s.t. $O(G) = 50$. then find the total number of generators for the given cyclic group $(G, *)$

- Positive integers that are less than 50 & Co-prime to 50 are,

1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39, 41, 43, 47, 49

There are 20 +ve integers less than 50 & Co-prime to 50.

∴ No. of generators of the above cyclic group = 20.



Topic : Properties w.r.t. cyclic group

- Let 'n' be any positive integer, then number of positive integers which are less than 'n' and coprime to 'n' can be given by Euler's totient function $\phi(n)$

$$\phi(n) = n * \left\{\left(1-\frac{1}{P_1}\right)\cdot\left(1-\frac{1}{P_2}\right)\cdot\left(1-\frac{1}{P_3}\right)\cdot\cdots\right\}$$

where $P_1, P_2, P_3, \ldots$ are Prime factors of 'n'

Q: Let $n = 50$, then find the no. of positive integers which are less than 50 & Co-prime to 50.

Soln:- Number of such positive integers = $\phi(50)$

$50 = 2 * 25$

$= (2)^1 * (5)^2$

let Prime Factor 'P_1'

let prime Factor 'P_2'

$= 50 * \left\{\left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{5}\right)\right\}$

$= 50 \times \frac{1}{2} * \frac{4}{5}$

$= 20$ Ans



Topic : Properties w.r.t. cyclic group

- Let $(G, *)$ be a cyclic group of order "$O(G) = n$".

then,

Total No. of generators of the given cyclic group = No. of +ve integers less than 'n' and coprime to 'n' = Euler's totient function $\phi(n)$

Q:: Let $(G, *)$ be a cyclic group s.t. $O(G) = 250$
Find the total no. of generators of given cyclic group.

Soln Ans $= \phi(250)$

$250 = (5)^3 * 2$
p_1 p_2

$= 250 * \left\{\left(1-\frac{1}{5}\right) \cdot \left(1-\frac{1}{2}\right)\right.$

$= 250 * \frac{4}{5} * \frac{1}{2}$

$= 100$ Ans



Topic : Properties w.r.t. cyclic group

- Let $(G,*)$ be a cyclic group such that $O(G) = P$ (Prime No.) then total no. of generators of the cyclic group

$$(G,*) = (P-1)$$

{ As all nos from 1 to $(P-1)$ are Co prime to P }

P = Prime No.
∴ the only prime factor of P = P

$$= \phi(P) = P * \left\{\left(1-\frac{1}{P}\right)\right\}$$

$$= (P-1)$$



Topic : Some important properties

1. Every cyclic group is abelian group.

$(a * b) * c$

$a * (b * c)$

let 'a' is the generator of cyclic group
for two element $P, q \in$ group

$P * q = (a)^r * (a)^s$

'a' $(r+s)$ times

$= (a * a * a * \cdots - a) * (a * a * a * - - - * a)$

r times a $\quad$ s times a

$= (a * a * - - - - - - - - - * a) * (a * a * - - - * a)$

s time a $\quad$ r times a

$= a^s * a^r$

$= q * p$

Bcoz '*' is associative



Topic : Some important properties

2. Every group of prime order is cyclic.



Topic : Some important properties

3. Every group of prime order is abelian.

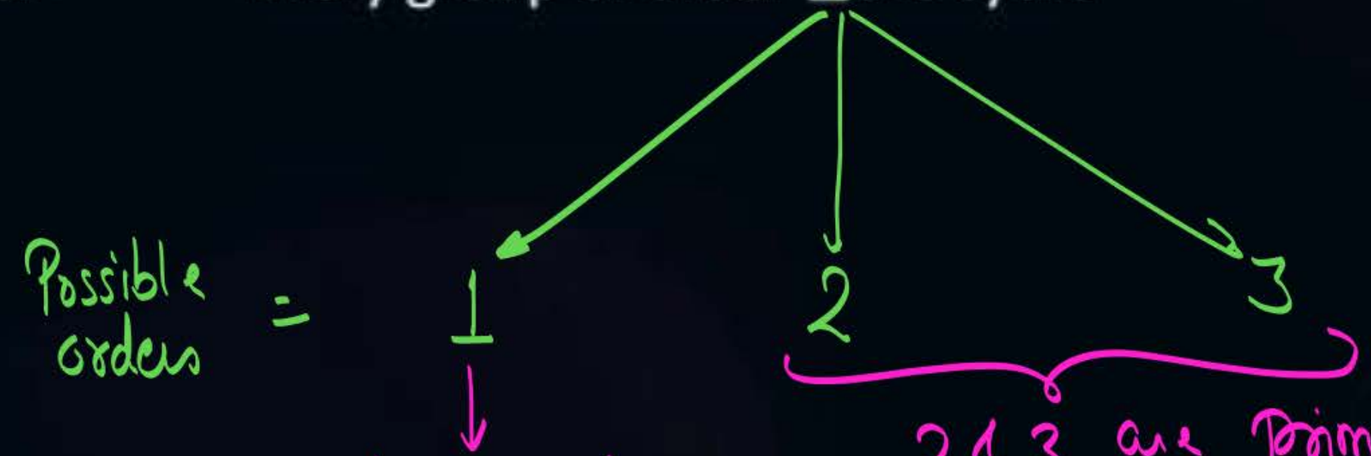
4. Every subgroup of a cyclic group is cyclic, but generator of the cyclic subgroup need not be same as the generator of the original cyclic group.

eg $\{1, -1, i, -i\}$ is a cyclic group w.r.t. multiplication.
and generators are i & $-i$

and $\{1, -1\}$ is sub-group of above group.
it is also a cyclic group with '-1' as generator

Topic : Some important properties

5. Every group of order ≤ 3 is cyclic





Topic : Some important properties

6. Every group of order ≤ 4 is cyclic except klein-4 group of order = 4

Properties of Klein-4 group :-

1. Order = 4
2. Every element must be inverse of itself
3. Operation on any two non-identity element must produce 3rd non-identity element

Eg: Set = {e, a, b, c} where e = identity
w.r.t opn '*'

1. order = 4
2. inv(e) = e
   inv(a) = a
   inv(b) = b
   inv(c) = c
3. a * b = c = b * a
   a * c = b = c * a
   b * c = a = c * b

{1, 3, 5, 7} w.r.t. $\otimes_8$ is Klein-4 type group.

1. order = 4
2. inv(1) = 1
   inv(3) = 3
   inv(5) = 5
   inv(7) = 7
3. $3 \otimes_8 5 = 7 = 5 \otimes_8 3$
   $3 \otimes_8 7 = 5 = 7 \otimes_8 3$
   $5 \otimes_8 7 = 3 = 7 \otimes_8 5$

'Klein-4' group is not Cyclic, but it is Commutative.



Topic : Some important properties

7. Klein-4 group is not cyclic group but it is an abelian group



Topic : Some important properties

8. Every group of order ≤ 5 except klein-4 group of order=4 is cyclic.



Topic : Some important properties

9. Every group of order ≤ 5 is abelian.



Topic : NOTE

1. In a group (G,*) if a*a = a, then a=identity element.

2. In a group (G,*) if a^{-1}=a ∀a∈G , then (G,*) is abelian.

3. In a group (G,*) if $(a * b)^2 = a^2 * b^2$, then (G,*) is abelian.

$a * a = a$

$(a*a)*a^{-1} = a*a^{-1}$

$a*(a*a^{-1}) = e$

$a * e = e$

$\boxed{a = e}$



Topic : NOTE

1. In a group (G,*) if a*a = a, then a=identity element.

2. In a group (G,*) if $a^{-1}=a$ $\forall a \in G$, then (G,*) is abelian, but Converse of statement need not be true

3. In a group (G,*) if $(a*b)^2 = a^2 * b^2$, then (G,*) is abelian.

Let $a,b \in (G,*)$ $\therefore (a*b) \in (G,*)$

$\therefore a^{-1}=a$

$b^{-1}=b$

$\therefore (a*b)^{-1} = a*b$

$b^{-1}*a^{-1} = a*b$

$\boxed{b*a = a*b}$ $\therefore$ Commutative

Eg: $(Z,+)$ is an abelian group bcz for elemen $a \in Z$

$inv(a) = -a \neq a$



Topic : NOTE

1. In a group (G,*) if a*a = a, then a=identity element.

2. In a group (G,*) if a^{-1}=a ∀a∈G , then (G,*) is abelian.

3. In a group (G,*) if $(a*b)^2 = a^2 * b^2$, then (G,*) is abelian.

$(a*b)^2 = a^2 * b^2$

$(a*b)*(a*b) = (a*a)*(b*b)$

$a*(b*a)*b = a*(a*b)*b$

$(b*a) = (a*b)$ Commutative

Law of cancellation w.r.t. group

if $a*b = a*c$ then $b = c$

but if $a*b = c*a$ then $b = c$ need not be true



Topic : NOTE

1. In a group (G,*) if a*a = a, then a=identity element.

2. In a group (G,*) if a^{-1}=a ∀a∈G , then (G,*) is abelian.

3. In a group (G,*) if $(a*b)^2 = a^2 * b^2$, then (G,*) is abelian.

$(a*b)^2 = a^2 * b^2$

$(a*b)*(a*b) = (a*a)*(b*b)$

$a*(b*a)*b = a*(a*b)*b$

$a^{-1}*a*(b*a)*b = a^{-1}*a*(a*b)*b$

$(b*a)*b*b^{-1} = (a*b)*b*b^{-1}$

Law of cancellation w.r.t. group

if $a*b = a*c$ then $b = c$

but if $a*b = c*a$ then $b = c$ need not be true

Can not be Cancelled



2 mins Summary

Topic Cyclic group

Topic Properties w.r.t. cyclic groups

Topic Some important properties

THANK - YOU

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